

Game DNA: Visual Analytics for Video Game Market Positioning

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Abstract. Game DNA is an interactive Visual Analytics system for exploring the multidimensional market positioning of PC video games on Steam. Starting from 125,855 Steam entries, a preprocessing pipeline produces a curated sample of 5,000 games represented through six dimensions: Popularity, Appreciation, Engagement, Commercial Position, Maturity, and Market Reach. Strongly skewed variables are log-transformed and all dimensions are standardized before Principal Component Analysis (PCA). The first two components explain 53.83% of the variance and form an interactive Game Map, while the complete six-dimensional representation is preserved in a Parallel Coordinates view. Coordinated brushing, highlighting, and selection support an overview–focus–refine workflow. Three computations are triggered by interaction: Group Characterization, Tag Enrichment, and Game Similarity. The system supports the discovery of market segments and comparable games, including similarities that cross conventional genre boundaries.

Keywords: Visual Analytics · Video Games · Steam · Market Analysis · Dimensionality Reduction

1 Introduction

Steam contains a large and heterogeneous collection of PC games that differ in popularity, player appreciation, engagement, price, age, and international reach. Considering these characteristics independently does not directly show how titles relate to each other when multiple market dimensions are considered simultaneously.

Game DNA addresses this problem through an interactive Visual Analytics system designed for a game-market analyst working for a small or medium-sized PC developer or publisher. The system supports the identification of market segments, the interpretation of the characteristics that distinguish them, and the discovery of comparable products. It is intended for exploratory decision support rather than for predicting commercial success.

Each title is represented through six dimensions: *Popularity* (number of reviews), *Appreciation* (positive-review ratio), *Engagement* (average playtime), *Commercial Position* (base price), *Maturity* (years since release), and *Market Reach* (number of supported languages). The analysis focuses on four questions:

which multidimensional groups emerge, which dimensions distinguish a selected group from the rest of the market, which titles have the most similar Game DNA, and whether similar market profiles can occur across different genres.

2 Related Works

Game DNA combines ideas from game analytics, dimensionality reduction, high-dimensional visualization, and coordinated multiple views. PLATO [5] demonstrates how Visual Analytics can support game analysis by combining interactive visual representations with analytical methods, although its focus is gameplay telemetry rather than comparison between games at market level. iPCA [2] shows how PCA can be integrated with coordinated views to connect a reduced projection with the original multivariate data. Game DNA follows this principle by using PCA as an overview while retaining the complete six-dimensional representation for detailed analysis.

The coordination between the PCA map and Parallel Coordinates is also related to the dual item- and dimension-oriented exploration described by Turkay et al. [4]. More generally, Roberts [3] discusses coordinated multiple views as a way to combine complementary representations of the same data. Game DNA applies these established principles to a market-oriented workflow in which visual selection directly triggers domain-specific computations such as group characterization, tag enrichment, and game similarity.

3 Methodology

3.1 Dataset and Game DNA Features

The project uses the *Steam Games Dataset* provided by FronkonGames [1], containing 125,855 entries. The raw collection also includes demos, playtests, software, servers, soundtracks, and records with insufficient information. Entries were therefore retained only when they had a valid release date, meaningful owner information, at least 100 reviews, positive average playtime, and at least one relevant game genre; auxiliary products were excluded. This reduced the dataset to 18,170 candidate games.

A proportional stratified random sample was then drawn using four popularity tiers based on estimated owners and four release eras, creating 16 strata. Using random seed 2025, the final sample contains 5,000 unique games and no missing values in the six Game DNA dimensions. With six analytical dimensions, the Angelini–Santucci index is $5000 \times 6 = 30,000$.

Popularity is computed as positive plus negative reviews, Appreciation as the ratio of positive reviews, and Maturity from a fixed reference date of 17 August 2026. When a temporary discount is present, the undiscounted base price is reconstructed. The four strongly right-skewed variables shown in Table 1 are transformed with $\log(1 + x)$; all six dimensions are then standardized with

$$z = \frac{x - \mu}{\sigma}.$$

Table 1. Game DNA dimensions and preprocessing.

Dimension	Feature	Preparation
Popularity	Total Reviews	$\log(1 + x)$
Appreciation	Positive Ratio	unchanged
Engagement	Avg. Playtime	$\log(1 + x)$
Commercial Position	Base Price	$\log(1 + x)$
Maturity	Release Age	unchanged
Market Reach	Language Count	$\log(1 + x)$

This prevents features with larger numerical scales from dominating the subsequent analysis.

3.2 Dimensionality Reduction

PCA is applied to the six standardized dimensions. PC1 explains 33.83% of the variance and PC2 20.00%, for a cumulative 53.83%. Their loadings suggest an interpretable market structure: PC1 is positively associated with Popularity, Engagement, Commercial Position, Market Reach, and Appreciation, and is therefore labelled *Market Presence*; PC2 is dominated by Maturity and is labelled *Market Legacy*. These labels are interpretative, and the sign of PCA components is arbitrary.

The first two components are used only as a two-dimensional overview. Because 46.17% of the variance remains outside this projection, the complete six-dimensional standardized representation is preserved for Parallel Coordinates and Game Similarity.

3.3 Interaction-Driven Analytics

The active analytical group follows the most specific current interaction: a Game DNA refinement has priority, followed by a locked Market Focus and finally a temporary Game Map brush. Three computations are then derived from the current interaction context.

For each dimension j , Group Characterization compares the selected group with all remaining games:

$$\Delta_j = \bar{z}_{selected,j} - \bar{z}_{rest,j}.$$

Tag Enrichment compares the prevalence of tag t in the two groups:

$$E_t = p(t \mid selected) - p(t \mid rest),$$

reported in percentage points and shown only for groups containing at least 10 games. For individual titles, Game Similarity uses Euclidean distance in the

complete standardized six-dimensional space:

$$d(A, B) = \sqrt{\sum_{j=1}^6 (z_{A,j} - z_{B,j})^2}.$$

The selected game is excluded and the five nearest titles among the remaining 4,999 games are returned. Genre is deliberately excluded from this distance.

4 Visualizations and Interaction

The frontend is implemented with React, Vite, and D3.js and combines four coordinated views. The *Game Map* represents all 5,000 titles in the PC1–PC2 space, with color encoding a simplified DisplayGenre category. The *Game DNA View* uses Parallel Coordinates to expose all six standardized dimensions. The *Analytics View* alternates between Group Characterization and Game Similarity, while the *Tag Analysis View* shows the five most positively and negatively enriched Steam tags for the current group.

The main workflow follows an *overview–focus–refine* strategy. A rectangular brush on the Game Map defines a temporary group and immediately updates Group Characterization and Tag Enrichment. The region can then be locked as a *Market Focus*; its PCA boundaries remain stable while the analyst refines the group through one or more brushes on the Parallel Coordinates axes. Multiple dimensional brushes are combined with logical AND. Search and genre filtering can be applied without destroying the locked focus or active refinement.

Coordination also works at individual-game level. Hovering a game in either the Game Map or Game DNA View highlights the corresponding representation in the other view. Clicking creates a persistent shared selection and switches the Analytics View to Game Similarity, while the Tag Analysis View continues to preserve the broader group context. Deselecting the game restores the previous Group Characterization.

5 Discovered Insights

The completed system was used as an exploratory environment rather than to confirm predefined conclusions. Four representative investigations are summarized below.

High presence and Free-to-Play refinement. A Market Focus in the upper-right PCA region contained 232 games and showed particularly high Popularity (+2.40) and Engagement (+1.64) relative to the remaining market. Multiplayer (+42.5 pp), Open World (+28.3 pp), and Co-op (+26.2 pp) were strongly enriched. Refining the same region to games with Commercial Position at or below the global mean reduced the group to 37 titles. Commercial Position became strongly negative (−1.60), while Popularity (+2.37) and Engagement (+1.98) remained very

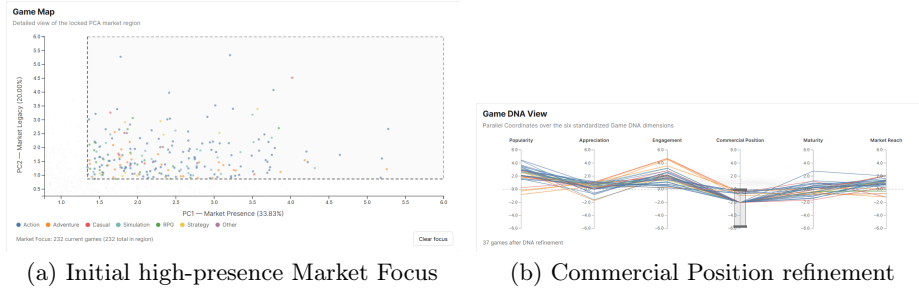


Fig. 1. Progressive exploration from a PCA market region to a more specific 37-game subsegment.

high. Free to Play (+60.5 pp), Multiplayer (+48.7 pp), and PvP (+39.3 pp) became the most distinctive tags. This demonstrates how a broad spatial pattern can be progressively refined into a more specific market subsegment.

Recent games with strong reach. A lower-right PCA region contained 149 games characterized by much lower Maturity (−1.32), but higher Market Reach (+1.21) and Commercial Position (+0.91). Popularity was only slightly above the reference market (+0.12). The segment was enriched in 3D, Exploration, Colorful, Singleplayer, and Stylized tags. The result identifies a recent market profile with broad reach and relatively high commercial positioning without the accumulated popularity of older high-legacy titles.

Cross-genre market similarity. From a mixed-genre Market Focus of 63 games, the analyst selected *Northgard: Definitive Edition*. Similarity was then computed against all other 4,999 games in the full six-dimensional space. The nearest titles were *Jurassic World Evolution* (0.435), *Warhammer: Vermintide 2* (0.547), *My Time at Portia* (0.559), *Mortal Kombat 11* (0.566), and *Dishonored 2* (0.631). Four of the five belong to a different DisplayGenre, showing that genre and multidimensional market similarity capture different properties.

High appreciation with low market presence. A low-Market-Presence region was refined through the Appreciation axis, producing 51 highly appreciated games. Appreciation was substantially above the reference market (+1.11), while Popularity (−0.88), Engagement (−1.69), and Market Reach (−0.79) were lower. Cute (+24.8 pp), Casual (+17.6 pp), Stylized (+16.7 pp), Colorful (+16.2 pp), and 2D (+14.0 pp) were over-represented. The result shows why positive user reception should not be interpreted as equivalent to broad visibility or high engagement.

These findings are descriptive associations within the analysed sample and should not be interpreted as causal evidence.

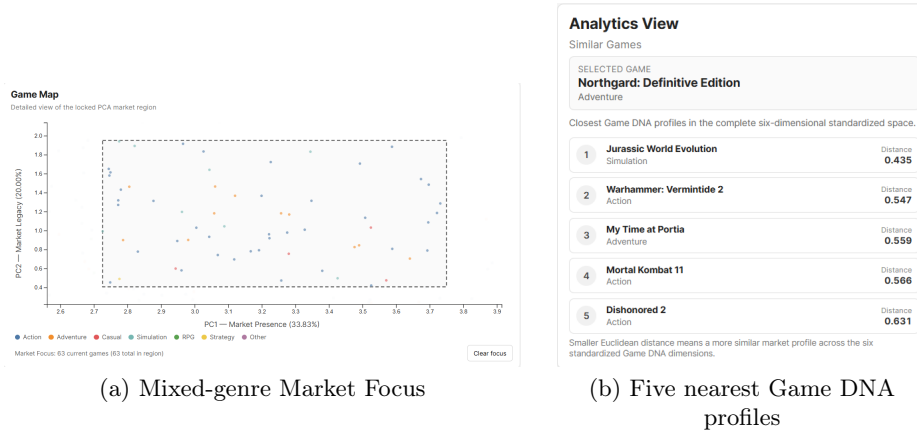


Fig. 2. Cross-genre similarity for *Northgard: Definitive Edition*.

6 Conclusions

Game DNA combines a PCA overview, the complete six-dimensional Game DNA representation, coordinated interaction, and interaction-driven analytics to support progressive exploration of the Steam market. The investigations show how analysts can move from global patterns to refined market segments and individual comparable titles, including similarities that cross genre boundaries.

The six dimensions remain proxies for market positioning and exclude factors such as revenue, development cost, marketing expenditure, and temporal evolution. Moreover, the PCA map preserves 53.83% of the variance, the group and tag analyses are descriptive rather than causal, and similarity assumes equal feature weights. Future work could incorporate temporal and commercial data, alternative similarity metrics, and evaluation with market analysts.

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